Amirmohammad Mohammadi
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Google Scholar | LinkedIn | Website | U.S. Permanent Resident

TECHNICAL FOCUS AREAS

- **Machine Learning & AI:** Data Analysis, Deep Learning, Foundation Models, Efficient Adaptation, Computer Vision, Multimodal Learning, Physics-Informed Neural Networks, Generative Modeling.

EDUCATION

Texas A&M University, College Station, Texas — September 2022 - December 2026
Doctor of Philosophy in Computer Engineering

Sharif University of Technology, Tehran, Iran — September 2018 - February 2021
Master of Science in Electrical Engineering

University of Tabriz, Tabriz, Iran — September 2014 - September 2018
Bachelor of Science in Electrical Engineering

PROFESSIONAL EXPERIENCE

Texas A&M University, College Station, Texas — September 2022 – December 2026
Graduate Research Assistant

- **Developed** a novel distribution-guided parameter-efficient fine-tuning method using histogram layers for Transformer-based foundation models for waveform-based sensing applications.
- **Adapted** a computational pathology framework to investigate parameter-efficient fine-tuning of vision-language models for ethanol drinking status prediction from liver histology under few-shot setting.
- **Reduced** ground truth data requirements by a factor of 15 for physiological signals by developing and evaluating physics-informed neural networks for cuffless blood pressure measurement.
- **Conducted** research supported by fully funded assistantship from the National Institutes of Health and MIT Lincoln Laboratory across AI-enabled digital health & earth observation applications.

PEER-REVIEWED PUBLICATIONS

- **A. Mohammadi,** R. Khadka, K. A. Grant, M. L. Benton, and J. Peeples, "Predicting Ethanol Drinking Status from Nonhuman Primate Liver Histology: A Proof of Concept for Parameter-efficient Fine-tuning in Few-Shot Vision-Language Modeling," *IEEE International Conference on Bioinformatics and Biomedicine,* 2026. (Accepted)
- **A. Mohammadi**, D. Carreiro, A. Van Dine, and J. Peeples, "Histogram-based Parameter-efficient Tuning for Passive and Active Sonar Classification," *IGARSS 2026 - 2026 IEEE International Geoscience and Remote Sensing Symposium*, Washington DC, DC, USA, 2026, pp. 5031-5035. [link].
- J. Ritu, **A. Mohammadi,** D. Carreiro, A. V. Dine, and J. Peeples, "Structural and Statistical Audio Texture Knowledge Distillation for Acoustic Classification," in *IEEE Access*, vol. 14, pp. 145754-145771, 2026. [link]
- F. Orvati Nia, **A. Mohammadi**, S. Al Kharsa, P. Naikare, Z. Hampel-Arias, and J. Peeples, "Neighborhood Feature Pooling for Remote Sensing Image Classification," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) Workshops*, 2026, pp. 1393–1402. [link]

- **A. Mohammadi**, I. Masabarakiza, E. Barnes, D. Carreiro, A. Van Dine and J. Peeples, "Investigation of Time-Frequency Feature Combinations with Histogram Layer Time Delay Neural Networks," *OCEANS 2025 Brest*, BREST, France, 2025, pp. 1-6. [link]
- **A. Mohammadi**, T. Kelhe, D. Carreiro, A. Van Dine, and J. Peeples, "Cross-Domain Knowledge Transfer for Underwater Acoustic Classification Using Pre-trained Models," in *OCEANS 2025 Brest*, BREST, France, 2025, pp. 1-6. [link]
- K. Sel, **A. Mohammadi**, R. I. Pettigrew, and R. Jafari, "Physics-informed neural networks for modeling physiological time series for cuffless blood pressure estimation," *npj Digital Medicine*, vol. 6, no. 1, p. 110, 2023. [link]
- **A. Mohammadi**, M. Fakharzadeh, and B. Baraeinejad, "An Integrated Human Stress Detection Sensor Using Supervised Algorithms," in *IEEE Sensors Journal*, vol. 22, no. 8, pp. 8216-8223, 2022. [link]

MANUSCRIPTS UNDER REVIEW

- **A. Mohammadi,** J. Peeples, and A. Van Dine "Parameter-efficient Dual-encoder Architecture with Differentiable Choquet Integral Fusion for Underwater Acoustic Classification". [link]

ACADEMIC SERVICE

- Reviewer, IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP).
- Session chair, IEEE International Geoscience and Remote Sensing Symposium (IGARSS).
- Delivered guest lectures in Machine Learning for Electrical Engineering at Texas A&M University.
- Mentored undergraduate and graduate students on end-to-end machine learning & AI projects.